Benchmarking Machine Learning Architectures and Optimal Transport for Temporal Cell-State Predictions on Single-Cell RNA Sequencing Data

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Abstract

Computational biologists use single-cell RNA sequencing (scRNA-seq) to measure transcriptional states in detail. However, a challenge with this is predicting how cell populations change over time from earlier observations because scRNA-seq is a destructive measurement technique. In this study, we evaluated six computational models for their ability to forecast unseen future transcriptional states. These models include Ridge Regression, Multi-Layer Perceptron, XGBoost Regressor, Random Forest Regressor, Neural ODE, and Temporal-Token Transformer. We trained them on earlier time points from two datasets: hydrogen-peroxide-treated human fibroblasts and TGF-β-treated human MCF10A breast epithelial cells undergoing epithelial-to-mesenchymal transition (EMT). The architectures were evaluated by predicting a completely held-out future cell population. We evaluated these models using squared Pearson correlation (r^2) for endpoint-expression accuracy, with optimal transport pairings and random pairings to test whether optimal transport pairings improved forecasting ability. All six models outperformed the persistence baseline (r^2 between each cell's Day-0 profile and its terminal prediction) across both datasets, with Random Forest and XGBoost achieving the strongest performance on the 2000 highly variable genes in the fibroblast and EMT datasets, respectively. Performance varied across model architectures, and high endpoint accuracy didn't always correspond to greater preservation of cell-to-cell variability. The results of this study support the ability of models to forecast future transcriptional states from time-course scRNA-seq data, but the most effective modeling approaches may depend on the cell type and biological perturbation. Future work across additional datasets could help determine which modeling strategies more generally capture both future population states and cellular variability.

Introduction

Computational biologists quantitatively model cell trajectories over time to understand how cellular states change across dynamic biological processes. A major problem in modelling these transitions is that single-cell RNA sequencing (scRNA-seq) is a destructive measurement technique, so researchers cannot track a single living cell continuously across days in scRNA-seq data (Tanay & Regev, 2017). As a result, researchers must infer relationships between separate static snapshot populations, a problem further complicated by technical noise and data sparsity (Kiselev et al., 2019). Because scRNA-seq provides high-dimensional snapshots rather than continuous observations of individual cells, computational models are increasingly used to learn and predict temporal relationships between cellular states (Ding et al., 2022). Methods from the mathematical field of optimal transport can be used to estimate probabilistic correspondences between populations by minimizing transport cost (Schiebinger et al., 2019).

Cells undergo dynamic changes in state in response to a variety of biological and environmental perturbations. Cellular senescence is a complex biological process characterized by stable cell-cycle arrest in response to cellular stress, and it plays important roles in physiological and pathological processes (Rattanavirotkul et al., 2020). Senescence can be induced by multiple stressors such as oxidative stress, DNA damage, telomere shortening, and oncogenic signaling (Handy & Loscalzo, 2011). Specifically, oxidative stress is important because excessive reactive oxygen species (ROS) can damage cellular biomolecules and disrupt cellular function. Mitochondrial function is closely tied to the rate of ROS production and oxidative stress (Murphy & Hartley, 2018), and mitochondrial dysfunction contributes to many common disorders, including neurodegeneration, metabolic disease, heart failure, different cancers, and chronic respiratory diseases (Hsu et al., 2016; W. Zhou et al., 2021; Sorrentino et al., 2018). The impact of oxidative stress and mitochondrial dysfunction on these different cellular processes and diseases highlights the biological importance of understanding cellular responses to stress.

Machine learning models provide a way for learning complex patterns within high-dimensional single-cell transcriptomic data and predict how cellular states and populations change over time (Sha et al., 2023; Cen et al., 2026). Neural ODE-based approaches such as RNAForecaster have demonstrated that it is possible to forecast gene-expression dynamics through continuous-time modeling (Erbe et al., 2023), while scNODE combines Neural ODE dynamics with a generative latent representation to predict cell populations at unmeasured time

points (Zhang et al., 2024). PRESCIENT similarly to this uses longitudinal single-cell population measurements to learn dynamics and simulate future cellular states through a neural-network-parameterized stochastic differential equation (Neural SDE) model (Yeo et al., 2021).

We implemented a six-model benchmarking framework using Ridge Regression (Hoerl & Kennard, 1970), a multilayer perceptron (MLP) (Rumelhart et al., 1986), a Random Forest Regressor (Breiman, 2001), a Neural ODE (R. T. Q. Chen et al., 2018), an XGBoost Regressor (T. Chen & Guestrin, 2016), and a temporal-token transformer (Vaswani et al., 2017) to evaluate forward prediction in sparse single-cell time-course data. We tested whether machine learning approaches could learn temporal patterns from scRNA-seq data collected during earlier stages of biological perturbations. These perturbations are given in the form of a dataset involving human fibroblasts treated with hydrogen peroxide (Burnaevskiy et al., 2023) and a second involving TGF-β-treated human MCF10A breast epithelial cells undergoing an epithelial-to-mesenchymal transition (EMT) process (Paul et al., 2023). We used the earlier time points as the training data to predict a held-out terminal endpoint for both datasets.

We hypothesized that an MLP would achieve higher predictive performance than the other evaluated model classes because its nonlinear architecture can learn complex relationships within high-dimensional single-cell transcriptomic data while avoiding some of the overfitting risks associated with more complex models (Ge et al., 2025; Srivastava et al., 2014).

Methods

Datasets

Two time-course scRNA-seq datasets were obtained from the Gene Expression Omnibus (GEO). GSE223128 contains human IMR-90 fibroblasts exposed to hydrogen peroxide (H_2O_2), with samples collected from untreated control (day 0) and at 4 hours, 1 day, 2 days, 3 days, 4 days, and 7 days after treatment (Burnaevskiy et al., 2023).

GSE200981 contains human MCF10A breast epithelial cells treated with TGF-β, with samples collected from untreated control (day 0) and at 4 hours, 1 day, 3 days, 4 days, 5 days, 6 days, and 8 days after treatment (Paul et al., 2023). We used this second dataset to evaluate whether predictive performance generalized across a distinct cell type, perturbation, and temporal sampling scheme.

Data Preprocessing

We processed gene-expression data using Scanpy (version 1.11.5) (Wolf et al., 2018). Counts were normalized to a total of 10,000 counts per cell and log1p-transformed. We then selected the 2,000 most highly variable genes for analysis, and gene-expression values were z-score standardized before principal component analysis (PCA).

PCA dimensionality was selected independently for each model through a training-only sweep over 10, 25, 50, 100, 250, 500, 750, and 1,000 PCA components. In the fibroblast dataset, for each PCA dimensionality, models were trained using transitions from Day 0 to time points through Day 3 and evaluated by directly forecasting Day 4 from held-out Day-0 validation cells. Hyperparameter optimization was only performed within the training data, while the held-out Day-0 test cells and Day-7 endpoint remained excluded from PCA fitting, model training, and dimensionality selection. The optimal dimensionality for each model was chosen as the number of PCA components producing the highest top 100 HVGs Pearson r^2 on the Day-4 validation population. This optimization selected 750 components for Ridge Regression, 500 for MLP, 10 for XGBoost, 10 for Random Forest, 250 for Neural ODE, and 100 for the Temporal-Token Transformer.

The same training-only PCA dimensionality-selection procedure was also applied separately to the EMT dataset. For this dataset, T6 (Day 6) was used as the validation endpoint, while T7 (Day 8) remained excluded from PCA fitting, model training, and dimensionality selection. This framework ended up selecting 500 components for Ridge Regression, 500 for MLP, 500 for XGBoost, 10 for Random Forest, 50 for Neural ODE, and 50 for the Temporal-Token Transformer.

Optimal Transport Alignment

ScRNA-seq measures different cells at each time point, so there is no direct correspondence between individual cells across time. Using the Python Optimal Transport (POT) library (version 0.9.6.post1) (Flamary et al., 2021), optimal transport was applied between Day-0 control cells and each future training time point using squared Euclidean distance in PCA space as the transport cost. We chose to use balanced entropic Sinkhorn optimal transport to estimate probabilistic source-target correspondences. The transport-cost matrix was scaled so that its maximum value was 1 before Sinkhorn optimization, which used a maximum of 5,000 iterations and a convergence threshold of 10^{-9}. PCA was fitted using the pre-terminal cell

populations before the 80/20 Day-0 train-test split, while the terminal endpoint population was excluded from PCA fitting and used only for final evaluation.

The resulting transport relationships were converted into deterministic source-target pairings using real observed target cells, with a target-use cap limiting the number of cells that could be paired to a single target cell. Sinkhorn hyperparameters were selected through a sweep over regularization values of 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, and 1, and target-use caps of 1, 2, 3, and 5. Hyperparameter combinations were evaluated using the arithmetic mean of the top 100 HVGs' Pearson correlation r^2 for XGBoost and MLP predictions on the validation endpoint. XGBoost and MLP were used because they were two high-performing models in the PCA selection and required less computational memory. For the fibroblast dataset, this achieved a highest Pearson correlation r^2 at a regularization value of 10^{-4} and a target-use cap of 3.

This same hyperparameter optimization procedure was also applied to the EMT dataset using T6 (Day 6) as its validation endpoint. With this dataset, the Pearson r^2 was maximized with a Sinkhorn regularization value of 10^{-2} and a target-use cap of 2. The resulting aligned source-target pairs were used as training data for the forecasting models.

To maintain a uniform input dimensionality across different frameworks, lower-dimensional PCA representations were padded with zeros to match the maximum component ceiling. Because squared Euclidean distance was used as the transport cost, this padding preserved pairing transport costs and did not alter the Sinkhorn couplings or source-target pairings within each PCA dimensionality.

Model Training

Before training and testing models, the control group was split 80% for training and 20% for testing after optimal transport pairings were generated. The 80% training cells were used with their paired future target cells for model training, while the remaining 20% of control cells were held out for testing. Their predicted future gene expression was then averaged and compared with the actual endpoint population.

For the fibroblast dataset, six models were trained, tested, and compared: Ridge Regression, a regularized linear model with an alpha value of 10.0; a multilayer perceptron (MLP), which learns nonlinear relationships through fully connected layers, with two hidden layers of 128 neurons, GELU activation, 0.2 dropout, a learning rate of 0.003, and 120 training

epochs; Random Forest, which combines multiple decision trees, with 20 trees, a maximum depth of 4, and 0.5 maximum features; XGBoost, which builds boosted decision trees sequentially to reduce prediction error, with 30 estimators, a maximum depth of 3, a learning rate of 0.05, 0.8 subsampling, and 0.8 column subsampling; a Neural ODE, which models cell-state changes over time using a learned velocity function, with a hidden width of 128, 6 Euler integration steps, 60 training epochs, a batch size of 512, a learning rate of 0.002, and weight decay of 0.001; and a Transformer-based model, which uses self-attention to learn relationships among transcriptomic features and temporal information, with a token size of 5, model dimension of 32, 4 attention heads, 1 Transformer layer, a feedforward dimension of 128, 0.1 dropout, 30 training epochs, a batch size of 512, a learning rate of 0.002, and weight decay of 0.00001.

Each model received a vector input consisting of the PCA-dimensional representation of a source cell, along with a time variable representing the change in time between the source and target time points. The model then predicted the future PCA representation of that cell.

Models were trained only on earlier time points. The terminal end-point was excluded from preprocessing fitting, optimal transport, and model training, and was used only for final evaluation.

The same six model architectures were applied to the EMT dataset using the same overall training and evaluation framework and the same individual model parameters

Forward Prediction

For the fibroblast dataset, models were trained using target time points through Day 4. The held-out Day-0 control cells were then used to directly predict the unseen Day 7 population using a seven-day time interval.

For the EMT dataset, models were trained using target time points through Day 6. The held-out Day-0 control cells were then used to directly predict the unseen Day 8 population using an eight-day time interval.

Model Evaluation

We evaluated the model performance for both the fibroblast and the EMT dataset using Pearson correlation r^2, calculated by averaging expression across cells for each gene and

computing the squared Pearson correlation between the observed and predicted gene-level mean-expression profiles. This metric was calculated across all 2,000 highly variable genes (HVGs) and for just the top 100 HVGs. The top five HVGs were additionally highlighted and named as individually colored points in model evaluation plots.

Mean-expression accuracy doesn't fully capture whether models preserve cell-to-cell heterogeneity, so we also chose to evaluate predicted cell-population distributions. We did this first by observing and predicting cell populations using the first two principal components to assess differences in population structure, and second, we calculated the 2-Wasserstein distance (W_2), a measure of how much the predicted cell population would need to shift to match the observed population, to quantify overall distributional discrepancy across varying PCA dimensions. W_2 was calculated instead of cell-to-cell variance because variance only measures the spread of the data, while W_2 measures the entire distribution simultaneously. Each of the six models was independently trained and evaluated using PCA dimensionalities of 2, 5, 10, 20, 50, 100, 200, 250, and 500. Ridge Regression and MLP were additionally evaluated at 750 dimensions, while the remaining four models were limited to 500 dimensions due to computational and memory constraints. A positive control was calculated by randomly dividing the terminal endpoint cells into two equal subsets and then computing the W_2 distance between them. A negative control was calculated by randomly sampling control cells and then comparing their distribution with the terminal endpoint distribution. For consistency, the number of cells sampled for all W_2 comparisons, including the positive and negative controls, and the model predictions, was kept equal to the number of cells in half of the endpoint population in both datasets.

Random-Pairing Control

For both datasets, a random-pairing control was used to compare optimal transport against random cell matching. The OT and random-pairing conditions used the same source-cell splits, model configurations, training procedures, and evaluation endpoints. They only differed in the source-target pairing procedure. We randomly paired the same source cells used for optimal transport with real observed cells from the corresponding training time points, and trained the models on these random pairs. The random pairings used the same target-use cap as the optimal transport pairings. We then compared performance from random pairing with performance from optimal transport pairings to determine whether OT produced more meaningful predictions of future cellular states.

Results

PCA Train-Only Sweep:

PCA dimensionality was optimized using training-only Day-4 validation across Ridge Regression, MLP, XGBoost, Neural ODE, Time Series Transformer, and Random Forest Regressor. PCA dimensionality was selected independently for each model based on the Pearson r^2 of the top 100 HVGs on the Day-4 validation population. The optimal dimensionalities were found to be 750 components for Ridge Regression (r^2 of 0.6860), 500 for MLP (r^2 of 0.7949), 10 for XGBoost (r^2 of 0.9012), 10 for Random Forest (r^2 of 0.8914), 250 for Neural ODE (r^2 of 0.6686), and 100 for the Temporal-Token Transformer (r^2 of 0.8952) (Fig. 1). Performance was relatively stable across many PCA dimensionalities for several models, although the optimal PC dimension differed for each model architecture. These results indicate that increasing PCA dimensionality didn't always improve forecasting performance and support the use of specific PCA dimensionalities for each model rather than a single shared dimensionality.

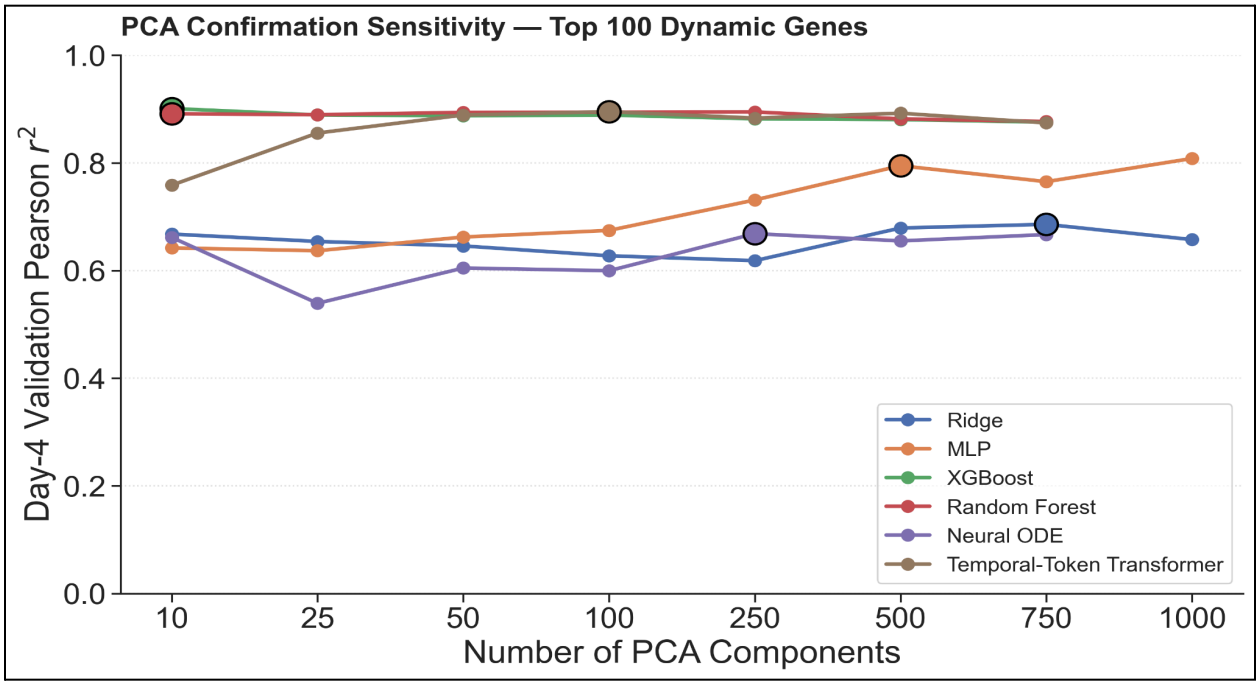


Figure 1. Training-Only PCA Dimensionality Selection and Model Sensitivity.
Day-4 validation Pearson r^2 across Ridge Regression, XGBoost, Random Forest Regressor, MLP, Neural ODE, and the Temporal-Token Transformer evaluated across PCA dimensionalities of 10, 25, 50, 100, 250, 500, 750, and 1,000 using the top 100 HVGs. Random Forest Regressor, Neural ODE, the Temporal-Token Transformer, and XGBoost were evaluated only

through 750 PCA dimensions due to computational memory constraints. Enlarged points indicate the PCA dimensionality selected for each model based on maximum top 100 HVG validation Pearson r^2. The figure was created using AI-assisted Python (Matplotlib) visualization code. All underlying data, analyses, and scientific interpretations were produced by the student researcher.

OT Hyperparameters Train-Only Sweep:

We evaluated OT hyperparameters using training-only Day-4 validation across XGBoost and MLP (Fig. 2). The optimal OT configuration used a Sinkhorn regularization value of 0.0001 and a target-use cap of 3, which were subsequently fixed for the final fibroblast dataset pipeline.

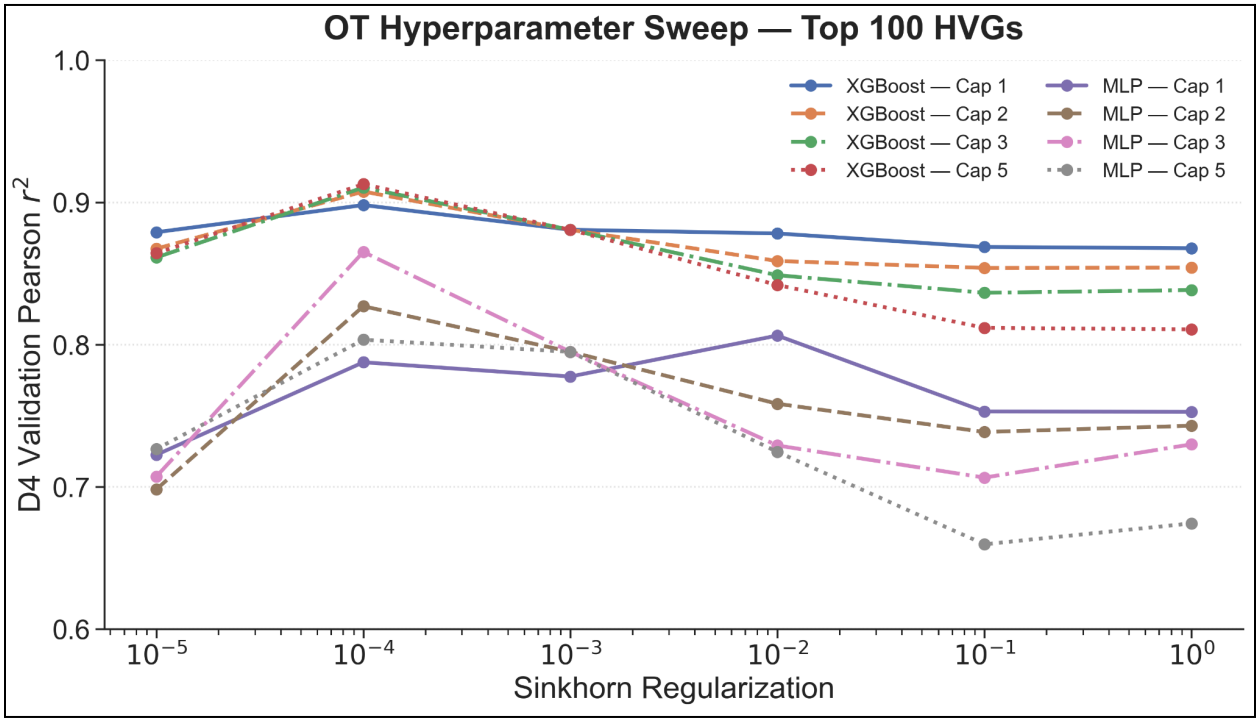


Figure 2. Training-only Sinkhorn optimal transport hyperparameter sweep.
Day-4 validation Pearson r^2 across XGBoost and MLP for top 100 highly variable genes across Sinkhorn regularization values of 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, and 1 and across target caps of 1, 2, 3, and 5. The figure was created using AI-assisted Python (Matplotlib) visualization code. All underlying data, analyses, and scientific interpretations were produced by the student researcher.

6 Model Benchmark Comparison:

Using the selected model-specific PCA dimensions and Sinkhorn optimal-transport settings, the six models were compared using Pearson r^2 across all 2,000 highly variable genes

and the top 100 highly variable genes under both optimal-transport and random-pairing conditions. Under optimal-transport pairing, Random Forest achieved the highest Pearson r^2 for both the full set of highly variable genes and the top 100 highly variable genes, with values of 0.931 and 0.818, respectively. Under random pairing, the Temporal-Token Transformer achieved the highest Pearson r^2 across all highly variable genes, with a value of 0.925, while Random Forest achieved the highest Pearson r^2 across the top 100 highly variable genes, with a value of 0.799. Optimal transport improved All-HVG Pearson r^2 for Ridge Regression, MLP, and Random Forest, but did not consistently outperform random pairing across all architectures.

Fibroblast Endpoint Expression-Profile Alignment					
Model	**PCA Dim.**	**Optimal Transport**		**Capped-Random**	
		All HVGs	**Top 100 HVGs**	**All HVGs**	**Top 100 HVGs**
Ridge Regression	750	0.9161	0.7533	0.9159	0.7629
Absolute MLP	500	0.9083	0.6907	0.9038	0.6860
XGBoost Regressor	10	0.9170	0.7686	0.9173	0.7713
Random Forest	10	**0.9314**	**0.8181**	0.9244	**0.7989**
Neural ODE	250	0.9016	0.6995	0.9238	0.7905
Temporal-Token Transformer	100	0.9225	0.7685	**0.9247**	0.7835
Day-0 Persistence	—	—	—	0.6049*	0.2723*

Figure 3. Six-model Endpoint Expression Comparison.

Day-7 mean expression prediction performance was evaluated using Pearson r^2 across Ridge Regression, MLP, XGBoost Regressor, Random Forest Regressor, Neural ODE, and Temporal-Token Time-Series Transformer. Each model was evaluated under four conditions: optimal transport pairing across all 2000 highly variable genes, optimal transport pairing across the top 100 highly variable genes, random pairing across all 2000 highly variable genes, and random pairing across the top 100 highly variable genes. The Day-0 persistence baseline for all 2000 highly variable genes and the top 100 highly variable genes are both shown, representing the prediction performance when the held-out Day-0 expression profile is used without any temporal change. The asterisk (*) indicates that the persistence baselines were only evaluated under capped-random pairing, not optimal transport. The table was formatted using AI-assisted LaTeX code. All underlying data, analyses, and scientific interpretations were produced by the student researcher.

Predicted vs Observed Expression for Ridge Regression and Random Forest

Random Forest tracked the observed gene-expression profile most closely across the six models with a Pearson r^2 value of 0.931 for all HVGs and 0.818 for the Top 100 Highly Variable Genes (Fig. 4B), while Ridge Regression, as the baseline model, achieved Pearson r^2 values of 0.916 and 0.753 for all HVGs and the top 100 highly variable genes (Fig. 4A), respectively. Pearson r^2 was higher across all 2,000 HVGs than across the top 100 highly variable genes for both models. Predictions from both models showed a balanced pattern of over- and under-prediction and were clustered around the perfect-agreement dashed line. For the five highlighted highly variable genes (ROBO2, FMN1, SERPINE1, MIR31HG, and SLC7A11), Random Forest exhibited an average absolute vertical distance of 0.282 from the perfect-agreement line, compared with 0.766 for Ridge Regression. Ridge Regression showed larger vertical distances for all five genes, particularly ROBO2 (1.253 versus 0.404) and SERPINE1 (1.188 versus 0.260). These expression-level patterns were consistent with the benchmark results in Figure 3, where Random Forest Regressor achieved the strongest All-HVG performance and strongest Top 100 HVG performance, while Ridge Regression performed worse than most other models in both categories.

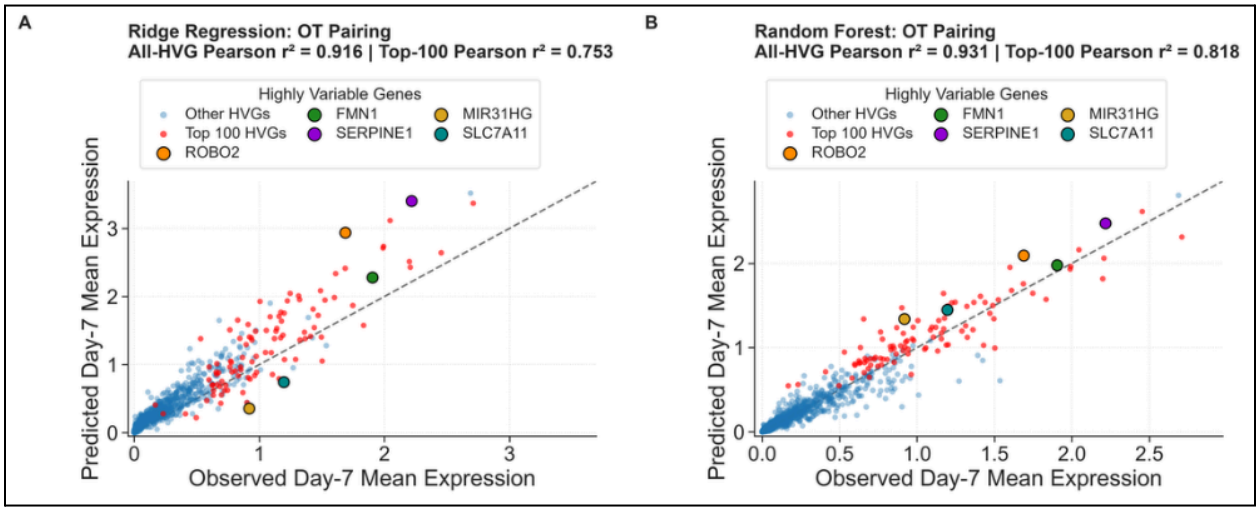


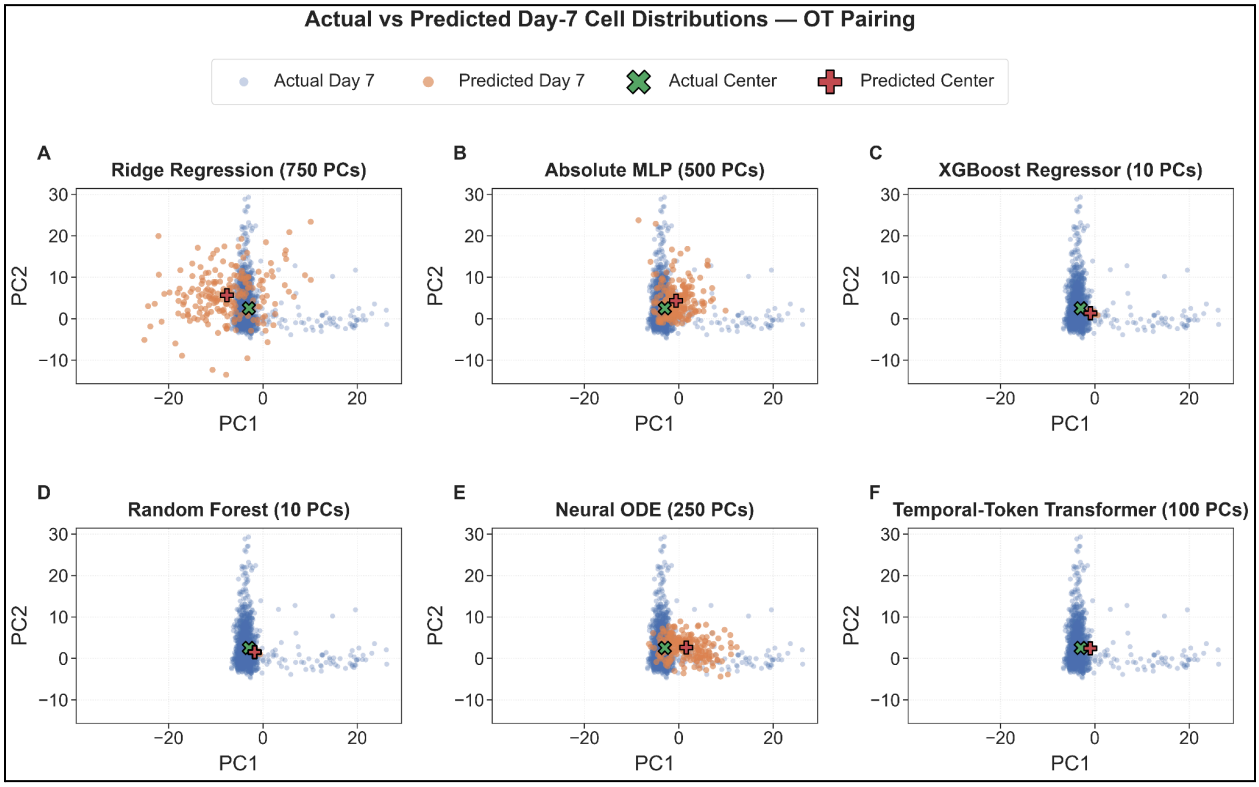
Figure 4. Observed Versus Predicted Day-7 Gene Expression.

Observed Day-7 mean gene expression was compared with predicted Day-7 mean expression for Ridge Regression (A) and Random Forest (B) using optimal transport-based training pairs for all 2000 highly variable genes (HVGs) and the top 100 HVGs. Each point represents one gene. The dashed diagonal line represents perfect agreement between observed and predicted mean gene expression; the blue dots represent all 2000 HVGs; the red dots represent the top 100 HVGs; and the five highlighted points identify the five most highly variable genes identified during training. The figure was created using AI-assisted Python (Matplotlib) visualization code.

All underlying data, analyses, and scientific interpretations were produced by the student researcher.

Cell-to-Cell Heterogeneity:

Predicted Day-7 cell-state distributions exhibited substantial differences in population structure across models (Fig. 5A-5F). In the first two principal components, Ridge Regression produced a broadly dispersed distribution shifted away from the observed population center, whereas XGBoost, Random Forest, and the Temporal-Token Transformer generated more concentrated predictions. At their originally selected PCA dimensionalities, W_2 distances were 44.03 for Ridge Regression at 750 PCs, 22.76 for MLP at 500 PCs, 9.24 for XGBoost at 10 PCs, 9.09 for Random Forest at 10 PCs, 23.85 for Neural ODE at 250 PCs, and 14.28 for the Temporal-Token Transformer at 100 PCs. Across 56 independently trained configurations, W_2 distance generally increased with PCA dimensionality. Among models evaluated at their originally selected PCA dimensionalities, Random Forest achieved the lowest reported W_2 distance of 9.09 at 10 PCs. This distance was approximately 1.58 times the positive-control distance of 5.76, and this was also 39.4% lower than the negative control distance of 15.00 (Fig. 5G).



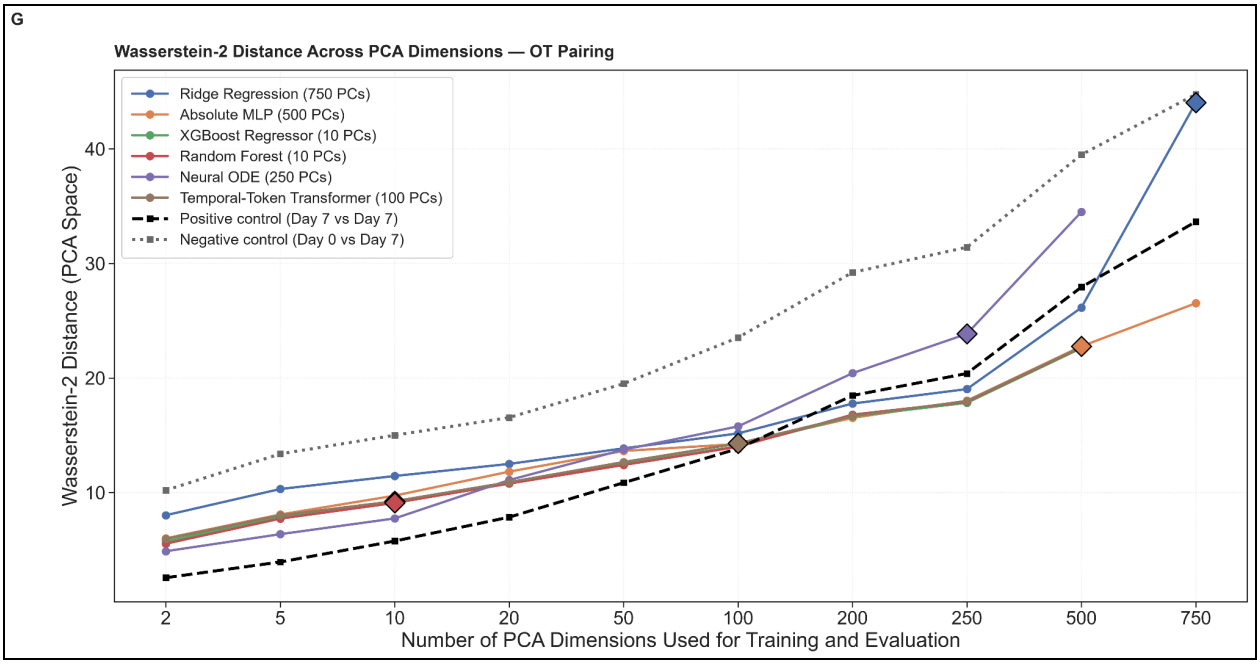


Figure 5. Preservation of Day-7 cell-state distributions and variance across forecasting models. (A-F) Observed and predicted Day-7 cell distributions under optimal-transport pairing projected onto the first two principal component dimensions. Each panel compares the observed Day-7 population with predictions from one of the six forecasting models using its optimized PCA dimensionality; green and red crosses indicate the observed and predicted population centers, respectively. **(G)** W_2 distance between predicted and observed Day-7 cell-state distributions across increasing PCA dimensionalities under optimal transport (OT) pairing. Lower values indicate greater similarity between predicted and observed distributions at a given dimensionality. Larger circular markers identify each model's originally selected PCA dimensionality. The maximum evaluated PCA dimensionality was limited to 500 components for XGBoost, Random Forest, Neural ODE, and the Temporal-Token Transformer due to computational memory constraints. A positive control is shown as the W_2 distance between independent samples of the Day-7 population, providing a reference for sampling variability. A negative control is shown as the W_2 distance between the initial and terminal cell populations, providing a baseline for comparison with model predictions. The figure was created using AI-assisted Python (Matplotlib) visualization code, but all underlying data, analyses, and scientific interpretations were produced by the student researcher.

EMT Dataset 6 Model Framework Comparison:

The six-model framework generalized to the independent EMT dataset, with all models outperforming the T0 (control) persistence baseline (the calculated Pearson r^2 between the predicted gene expression of the cell's control profile and its actual gene expression for the terminal endpoint). Under optimal-transport pairing, XGBoost achieved the highest All-HVG Pearson r^2 (0.974), while the Temporal-Token Transformer achieved the highest Top-100-HVG Pearson r^2 (0.944). As in the fibroblast dataset, the effect of optimal transport was architecture-dependent.

EMT Endpoint Expression-Profile Alignment					
Model	PCA Dim.	Optimal Transport		Capped-Random	
		All HVGs	Top 100 HVGs	All HVGs	Top 100 HVGs
Ridge Regression	500	0.9516	0.8878	0.9489	0.8838
Absolute MLP	500	0.9661	0.9292	0.9446	0.8896
XGBoost Regressor	500	**0.9737**	0.9419	0.9761	0.9517
Random Forest	10	0.9659	0.9182	0.9745	0.9479
Neural ODE	50	0.9562	0.9082	0.9361	0.8786
Temporal-Token Transformer	50	0.9724	**0.9436**	**0.9777**	**0.9573**
Day-0 Persistence	—	—	—	0.5316*	0.0656*

Figure 6. EMT Dataset Six-Model Endpoint Expression Comparison.
Day-8 mean expression prediction performance was evaluated using Pearson r^2 across all six models under the same four conditions used for the fibroblast dataset. The Day-0 persistence baseline for all 2000 highly variable genes and the top 100 highly variable genes are both shown similarly to the fibroblast table (Fig. 3), representing the prediction performance when the held-out T0 (control) expression profile is used without any temporal change. The asterisk (*) indicates that the persistence baselines were only evaluated under capped-random pairing, not optimal transport. The table was formatted using AI-assisted LaTeX code. All underlying data, analyses, and scientific interpretations were produced by the student researcher.

Predicted vs Observed Expression for XGBoost for EMT Dataset:

XGBoost closely tracked the observed gene-expression profile, achieving the highest Pearson r^2 of 0.974 across all HVGs, and second with 0.942 across the top 100 highly variable genes (Fig. 7). Predicted expression values generally clustered around the perfect-agreement line, with a balanced pattern of over- and under-prediction. For XGBoost under OT pairing, the five highlighted highly variable genes (KRT15, SERPINE1, KRT81, FN1, and KRT16) exhibited

an average absolute vertical distance of 0.289 from the perfect-agreement line, with individual distances ranging from 0.074 to 0.447. These findings complemented the strong overall endpoint mean-expression agreement.

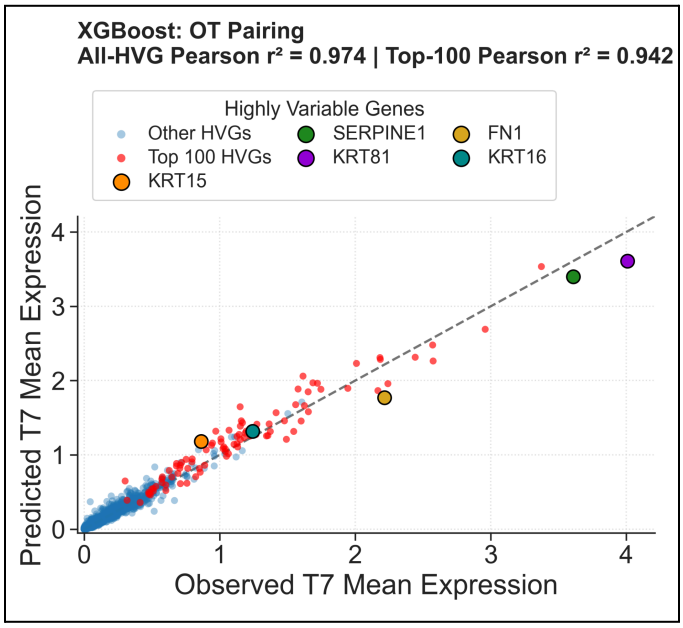
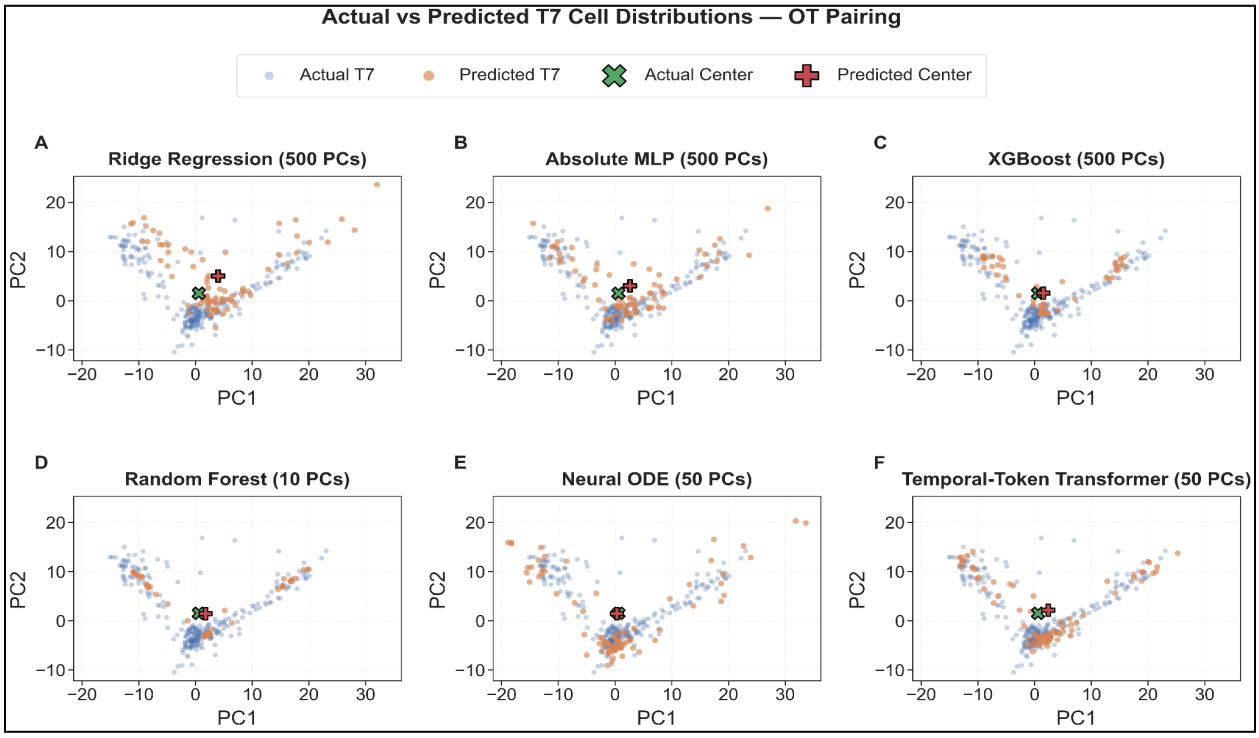


Figure 7. Observed Versus Predicted Day-8 Gene Expression.
Observed Day-8 mean gene expression was compared with XGBoost-predicted Day-8 mean expression using optimal transport-based training pairs under the same evaluation conditions as the fibroblast dataset (Fig. 4). Each point represents one gene. The dashed diagonal line represents perfect agreement between observed and predicted mean expression, and the five highlighted points identify the five most highly variable genes based on training-set variance. The figure was created using AI-assisted Python (Matplotlib) visualization code. All underlying data, analyses, and scientific interpretations were produced by the student researcher.

EMT Dataset Cell-to-Cell Heterogeneity

Predicted Day-8 cell-state distributions differed in their ability to reproduce the observed population structure (Fig. 8A). Although all six models captured portions of the branching pattern visible in the first two principal components, Ridge Regression and Neural ODE produced more dispersed predictions, whereas XGBoost and Random Forest generated more concentrated distributions. At their originally selected PCA dimensionalities, W_2 distances were 29.56 for Ridge Regression at 500 PC dimensions, 24.90 for MLP at 500 PC dimensions, 23.98 for

XGBoost at 500 PCs; 8.87 for Random Forest at 10 PCs; 15.05 for Neural ODE at 50 PCs, and 11.51 for the Temporal-Token Transformer at 50 PCs. Across the PCA sensitivity analysis, W_2 distance generally increased with dimensionality (Fig. 8B).



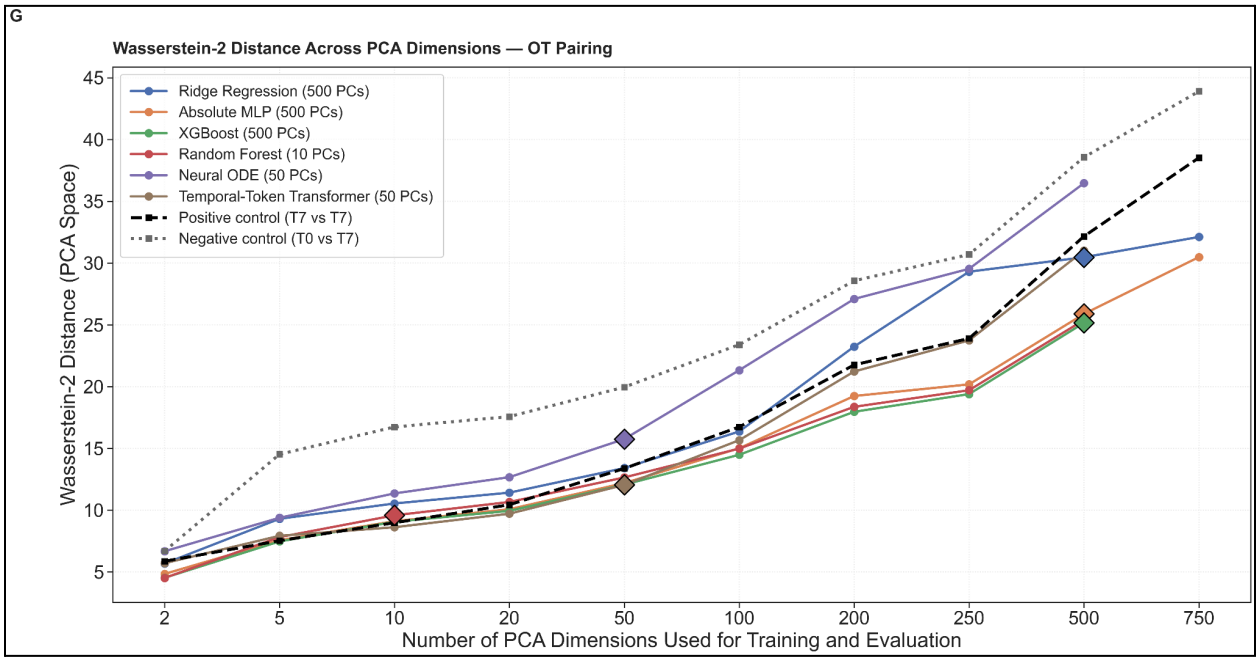


Figure 8. EMT Dataset Preservation of Day-8 Cell-State Distributions and Variance Across Forecasting Models. (A) Observed and predicted Day-8 cell-state distributions under the same conditions used for the fibroblast dataset. **(B)** W_2 distance between predicted and observed Day-8 cell-state distributions across increasing PCA dimensionalities under optimal transport (OT) pairing. The maximum evaluated PCA dimensionality was limited to 500 components for XGBoost, Random Forest, Neural ODE, and the Temporal-Token Transformer due to computational memory constraints. A positive control is shown as the W_2 distance between independent samples of the T7 (Day 8) population, providing a reference for sampling variability. A negative control is shown as the W_2 distance between the initial and terminal cell populations, providing a baseline for comparison with model predictions. The figure was created using AI-assisted Python (Matplotlib) visualization code. All underlying data, analyses, and scientific interpretations were produced by the student researcher.

Discussion

Predictability of Future Transcriptional States

This study demonstrates that future transcriptional states can be predicted from time-course single-cell RNA sequencing data using machine learning models. Across the fibroblast dataset, all six models exceeded the Day-0 persistence baseline when predicting the held-out Day-7 population across 2,000 HVGs and the top 100 HVGs. This indicates that the models learned temporal patterns in gene expression beyond simply preserving the initial transcriptional state.

Several models' ability to predict the held-out endpoint suggests that time-course transcriptomic data contain signals that multiple modeling strategies can capture. Random Forest achieved the strongest endpoint performance under optimal-transport pairing in the fibroblast dataset, despite having much fewer parameters than the Neural ODE and Temporal-Token Transformer. Increasing model complexity, through more parameters, therefore did not consistently correlate with greater endpoint forecasting accuracy in this system. The independent EMT dataset provided further evidence that the forecasting framework was not limited to fibroblast senescence. All six models again substantially exceeded the T0 persistence baseline when forecasting the held-out Day-8 population, with XGBoost and the Temporal-Token Transformer achieving strongest endpoint performance.

These findings are consistent with previous work demonstrating that temporal information in single-cell population snapshots can be used to predict future transcriptional

states. Previous approaches, including RNAForecaster, scNODE, and PRESCIENT, have demonstrated the potential of neural differential equation models for predicting future cellular states from single-cell transcriptomic data (Erbe et al., 2023; Zhang et al., 2024; Yeo et al., 2021). The present results extend this general observation by showing that temporal information can be used to predict future cell states by several simple regression and tree-based architectures under a common forecasting framework. The strong performance of tree-based models suggests that non-neural network ML methods can make comparably accurate gene expression predictions in the evaluated datasets to neural network ML methods.

The present study evaluated two biological systems: oxidative stress-associated fibroblast senescence and TGF-β-induced epithelial-to-mesenchymal transition. These systems differ biologically, but they cannot represent the full range of transcriptional dynamics present across different stressors and perturbation processes. Forecasting difficulty may also depend on factors such as the number and spacing of sampled time points, cell number, transcriptional variability, and the complexity of the underlying biological transition. Additionally, successful prediction of a held-out endpoint does not show that the same models would extrapolate equally as well over longer time intervals or across different cell types and perturbations.

Future work can evaluate this six-model framework across more longitudinal single-cell datasets representing a larger variety of biological processes and experimental time sampling designs. Testing prediction performance while systematically varying the number of observed time points, forecast horizon, and sampling density could help determine which properties of a time-course dataset could make future transcriptional states more predictable. Such experiments would further establish whether the strong performance of architectures with fewer parameters persists across more complex biological transitions or whether larger time-based models become advantageous as prediction difficulty increases.

Optimal Transport Does Not Universally Improve Forecasting

The comparison between optimal transport and random pairing shows that the benefit of inferred pairings depends on the downstream model architecture. Optimal transport improved endpoint performance for some model architectures, while others achieved similar or stronger performance with random pairing. This was also seen for predictions on the EMT dataset, where different architectures than those in the fibroblast dataset benefited from optimal transport. Thus, more structured cell pairings didn’t consistently mean improved extrapolation to an unseen future population.

Optimal transport has previously been used to address the lack of direct cell correspondences in time-course scRNA-seq data. Waddington-OT used sequential population snapshots to infer pairings and reconstruct developmental trajectories (Schiebinger et al., 2019), while CellOT applied optimal-transport principles to model responses between unpaired cell populations (Bunne et al., 2023). The present results don't challenge the ability of optimal transport to identify pairings between populations, but instead suggest that the benefit can depend on how they are used by a particular modelling pipeline.

A limitation of this comparison is that only one optimal transport framework, balanced entropic Sinkhorn transport with capped deterministic assignment, was evaluated against a capped-random control. Other transport methods of incorporating the full probabilistic coupling could produce different results. Future work could therefore examine whether other OT formulation strategies or training objectives improve model use of inferred cell correspondences for future-state prediction.

Endpoint Accuracy and Cell-to-Cell Heterogeneity Capture Different Properties

Higher endpoint mean-expression accuracy through Pearson r^2 did not necessarily correspond to better preservation of predicted cell-population distributions. In the fibroblast dataset, Random Forest, XGBoost, and the Temporal-Token Transformer achieved strong mean-expression agreement while producing substantially more concentrated predicted populations than those observed at Day 7 in the PC1–PC2 visualizations. In contrast, Ridge Regression generated a more dispersed predicted population structure, while its distribution was shifted relative to the observed cells. Neural ODE also produced a broader distribution, demonstrating that models differed in their ability to reproduce population structure despite achieving relatively strong endpoint-expression accuracy. Similar differences were observed in the EMT dataset, where models varied in the spread and positioning of their predicted cell populations.

These findings highlight the importance of evaluating distributional agreement alongside endpoint Pearson r^2. Although Pearson r^2 measures agreement between observed and predicted mean gene-expression profiles, it does not capture how accurately a model reproduces the distribution of individual cell states. W_2 distance provides an additional measure by quantifying the discrepancy between observed and predicted cell populations in PCA space. Across both datasets, the PCA sensitivity analyses revealed that W_2 distance generally increased as more principal components were included. This pattern indicates that the

dimensionality of the evaluation space substantially influences the measured distributional discrepancy, making direct comparisons between models evaluated at different PCA dimensionalities less informative. Previous single-cell prediction approaches, including CellOT, have similarly incorporated distributional evaluation, supporting the use of complementary metrics rather than relying exclusively on mean-expression accuracy (Bunne et al., 2023).

Nevertheless, neither the PC1–PC2 visualizations nor W_2 distances establish whether predicted variation corresponds to biologically meaningful cellular states. The visualizations represent only the first two principal components, while W_2 measures overall distributional discrepancy without identifying the biological processes responsible for differences between populations. Furthermore, a more dispersed predicted distribution does not necessarily indicate greater biological accuracy. Future work could incorporate cell-state proportions, gene-expression covariance, and established biological marker expression to determine whether predicted population structures reflect experimentally observed cellular heterogeneity.

Conclusion

We originally hypothesized that the MLP would achieve the highest predictive performance because its nonlinear architecture could learn complex relationships within high-dimensional single-cell transcriptomic data while avoiding some of the overfitting risks associated with more complex models. This hypothesis was not supported by the results. Although the MLP outperformed the persistence baseline, Random Forest achieved the strongest endpoint performance in the fibroblast dataset for 2000 HVGs under optimal transport, while XGBoost achieved the best performance in the EMT dataset for all 2000 HVGs under optimal transport.

One possible explanation for this result is that the relationships needed to predict endpoint expression may have been captured sufficiently by the tree-based models, which can also learn nonlinear relationships among PCA-derived features. Although the MLP can learn complex feature interactions through its hidden layers, this additional flexibility may not have provided an advantage for the prediction task. Differences in model complexity may also have influenced generalization, as models with insufficient capacity can underfit, while models with greater capacity may overfit the training data. The differing performance of the six architectures across the two datasets further suggests that model effectiveness depends on the biological system and prediction task rather than on a single architecture being universally superior.

A core limitation is that this benchmark evaluated a finite set of model configurations across only two biological systems. Although the architectures were compared within a common framework, more extensive model-specific hyperparameter optimization could change their relative performance. Future work could test these architectures across different situations as stated previously to confirm these findings.

Ultimately, the findings of this study demonstrate that machine learning approaches have the ability to forecast changes in cellular populations under biological perturbations, which provides a basis for future research into predicting cellular responses to oxidative stress and other biological perturbations.

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